

Lagrangian density

$$-\beta_3 \mathcal{A}^{\theta}_{, \mu} \mathcal{A}^{\mu}_{\theta} + \frac{1}{6} \chi \partial_{\theta} \mathcal{A}^{\nu}_{\mu} \partial^{\mu} \mathcal{A}^{\theta}_{, \nu} -$$

$$\frac{1}{6} \chi \partial_{\mu} \mathcal{A}^{\nu}_{\theta} \partial^{\mu} \mathcal{A}^{\theta}_{, \nu} + \frac{1}{9} \chi \partial^{\mu} \mathcal{A}^{\theta}_{, \nu} \partial_{\nu} \mathcal{A}^{\nu}_{\theta \mu} +$$

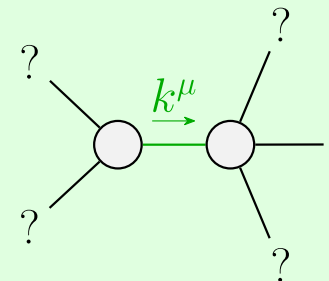
$$\frac{1}{18} \chi \partial^{\mu} \mathcal{A}^{\theta}_{, \nu} \partial_{\nu} \mathcal{A}^{\nu}_{\theta \mu} - \frac{1}{18} \chi \partial^{\mu} \mathcal{A}^{\theta}_{, \nu} \partial_{\nu} \mathcal{A}^{\nu}_{\theta \mu}$$

Added source term(s): $\mathcal{A}^{\alpha\beta\chi} \sigma_{\alpha\beta\chi}$

Source constraints	SO(3) irreps	#
$\mathcal{O}_0^{\#1\alpha\beta\chi} = 0$	$\mathcal{O}_0^{\#1\alpha\beta\chi}$	1
$\mathcal{O}_{1+}^{\#2\alpha\beta} = 0$	$\mathcal{O}_{1+}^{\#2\alpha\beta}$	3
$\mathcal{O}_{1+}^{\#1\alpha\beta} = 0$	$\mathcal{O}_{1+}^{\#1\alpha\beta}$	3
$\mathcal{O}_{2+}^{\#1\alpha\beta\chi} = 0$	$\mathcal{O}_{2+}^{\#1\alpha\beta\chi}$	5
$\mathcal{O}_{2+}^{\#1\alpha\beta} = 0$	$\mathcal{O}_{2+}^{\#1\alpha\beta}$	5
Total #:		17

	$\mathcal{A}_{2+}^{\#1\alpha\beta}$	$\mathcal{A}_{2+}^{\#1\alpha\beta\chi}$
$\mathcal{A}_{2+}^{\#1\alpha\beta}$	0	0
$\mathcal{A}_{2+}^{\#1\alpha\beta\chi}$	0	0

$$\text{Det}(1^-) \rightarrow -\frac{1}{288} \chi^2 k^4$$



Quadratic pole

Pole residue: $\frac{12}{\chi} + \frac{144\beta_3}{\chi^2 p^2} + \frac{p^2}{\beta_3} > 0$

Polarisations: 1

	$\mathcal{A}_{0+}^{\#1}$	$\mathcal{A}_{0+}^{\#1\alpha\beta\chi}$
$\mathcal{A}_{0+}^{\#1}$	$\frac{3\beta_3}{2}$	0
$\mathcal{A}_{0+}^{\#1\alpha\beta\chi}$	0	0

	$\sigma_{2+}^{\#1\alpha\beta}$	$\sigma_{2+}^{\#1\alpha\beta\chi}$
$\sigma_{2+}^{\#1\alpha\beta}$	0	0
$\sigma_{2+}^{\#1\alpha\beta\chi}$	0	0

	$\sigma_{1+}^{\#1\alpha\beta}$	$\sigma_{1+}^{\#2\alpha\beta}$	$\sigma_{1-}^{\#1\alpha}$	$\sigma_{1-}^{\#2\alpha}$
$\sigma_{1+}^{\#1\alpha\beta}$	0	0	0	0
$\sigma_{1+}^{\#2\alpha\beta}$	0	0	0	0
$\sigma_{1-}^{\#1\alpha}$	0	0	$\frac{\beta_3}{2 \text{Det}(1^-)}$	$\frac{12\beta_3 + \chi k^2}{12\sqrt{2} \text{Det}(1^-)}$
$\sigma_{1-}^{\#2\alpha}$	0	0	$\frac{12\beta_3 + \chi k^2}{12\sqrt{2} \text{Det}(1^-)}$	$\frac{\beta_3 + \frac{\chi k^2}{6}}{\text{Det}(1^-)}$

	$\mathcal{A}_{1+}^{\#1\alpha\beta}$	$\mathcal{A}_{1+}^{\#2\alpha\beta}$	$\mathcal{A}_{1-}^{\#1\alpha}$	$\mathcal{A}_{1-}^{\#2\alpha}$
$\mathcal{A}_{1+}^{\#1\alpha\beta}$	0	0	0	0
$\mathcal{A}_{1+}^{\#2\alpha\beta}$	0	0	0	0
$\mathcal{A}_{1-}^{\#1\alpha}$	0	0	$\beta_3 + \frac{\chi k^2}{6}$	$-\frac{12\beta_3 + \chi k^2}{12\sqrt{2}}$
$\mathcal{A}_{1-}^{\#2\alpha}$	0	0	$-\frac{12\beta_3 + \chi k^2}{12\sqrt{2}}$	$\frac{\beta_3}{2}$

	$\sigma_{0+}^{\#1\alpha\beta\chi}$	$\sigma_{0+}^{\#1}$	$\sigma_{0-}^{\#1\alpha\beta\chi}$
$\sigma_{0+}^{\#1\alpha\beta\chi}$	0	$\frac{3\beta_3}{2}$	0
$\sigma_{0+}^{\#1}$	0	0	0

Unitarity conditions

$$\chi < 0 \ \&\& \ \beta_3 > 0 \quad || \quad \chi > 0 \ \&\& \ \beta_3 > 0$$

(No massive particles)